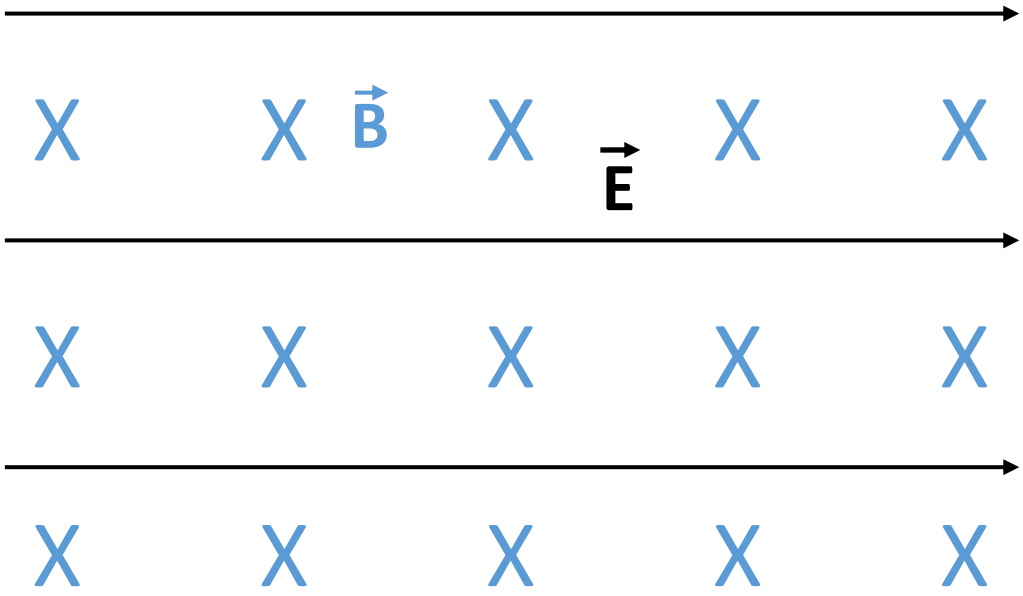
Exercise 1.

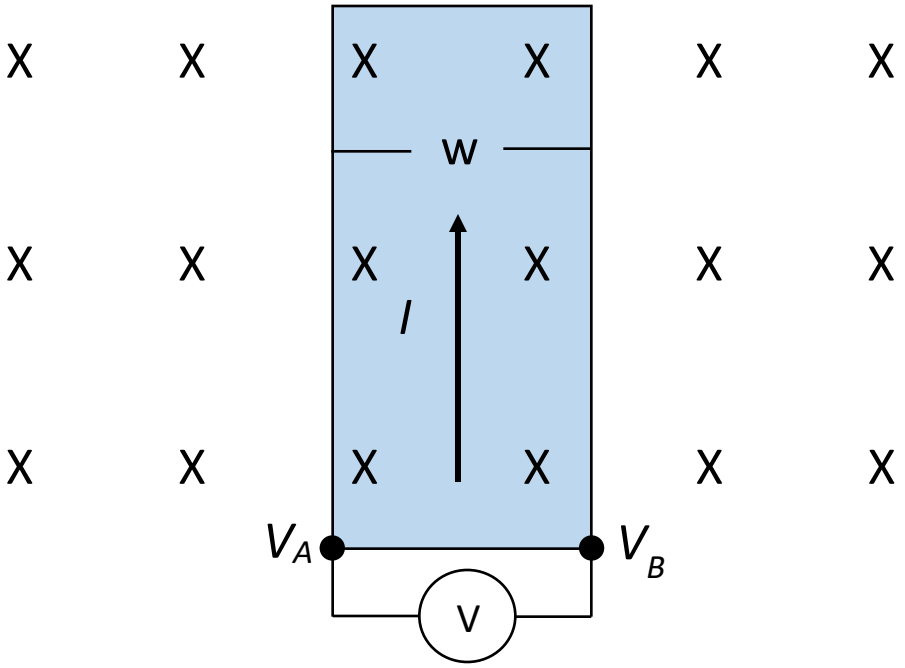


An electron with elementary charge $-e$ is first accelerated by a constant electric field $E_a = 2.5 \cdot 10^2$ V/m over a distance $L = 0.73$ m, after which it is observed to move in a straight line in the combined E- and B-fields shown in the figure.

a) Determine the speed of the electron after the acceleration phase.

b) If the electric field strength $E = 3.5 \cdot 10^4$ V/m, what is then the magnetic field strength B?

Exercise 2.



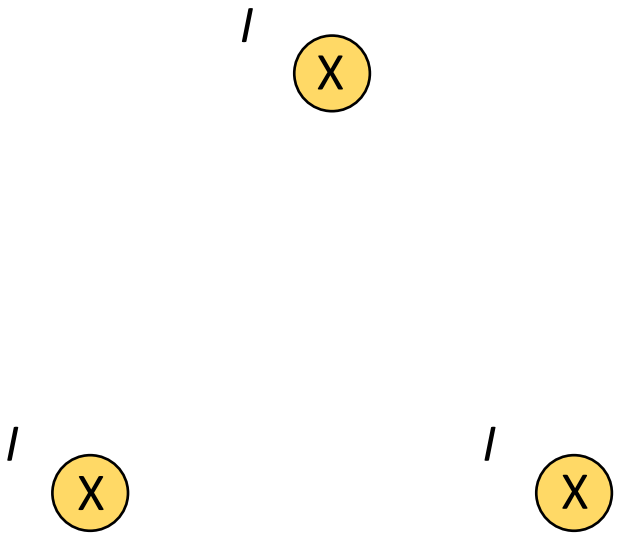
We attempt to measure the strength of a magnetic field using a Hall-sensor consisting of a semiconductor strip having a width w=4.2 cm and a thickness h=0.5 cm perpendicular to the plane of the figure. The magnetic field B points into the plane as shown, and has a field strength of $1.0 \cdot 10^{-4}$ T. The charge carrier density of the semiconductor is $n=3.5 \cdot 10^{21}$ m^{-3} and the charge carriers are positive elementary charges. We drive a current I=10.0 mA through the semiconductor directed upward in the figure as shown.

a) Determine the direction of the magnetic (Lorentz) force on the charge carriers in the semiconductor.

b) Determine the direction of the transverse electric field over the semiconductor needed to achieve equilibrium, *i.e.* that the sum of transverse electric and magnetic forces is zero. Does the potential difference V_B-V_A become positive or negative?

c) Determine an expression and a value for V_B-V_A as registered on the voltmeter.

Exercise 3.



Three long straight wires are arranged in an equilateral triangle as shown in the figure. In all three wires a current flows directed into the plane shown, and the magnitude of the current *I* is the same in all the wires. Where is the resultant magnetic force on the top wire directed?

A) Upwards

B) Downwards

C) Into the plane

D) Out of the plane

E) Towards the left

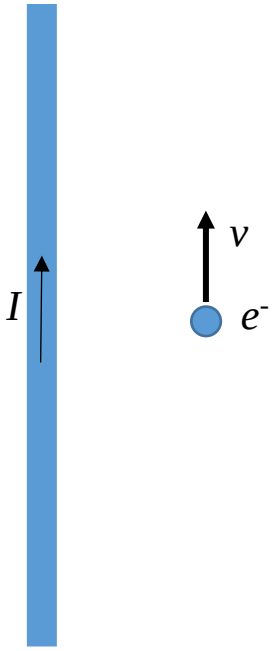
F) Towards the right

G) Another direction

H) The resultant force is zero.

I) Don't know

Exercise 4.



An electron moves with a speed of $3.5 \cdot 10^6$ m/s parallel to a wire carrying a current I=2.7 A, as shown in the figure. The distance from the electron to the center of the wire is 12 cm. Determine the magnitude and direction of the magnetic force on the electron from the wire.

A) $1.3 \cdot 10^{-18}$ N, towards the right

B) $2.5 \cdot 10^{-18}$ N, towards the right

C) $5.0 \cdot 10^{-18}$ N, towards the right

D) $7.4 \cdot 10^{-18}$ N, towards the right

E) $1.3 \cdot 10^{-18}$ N, towards the left

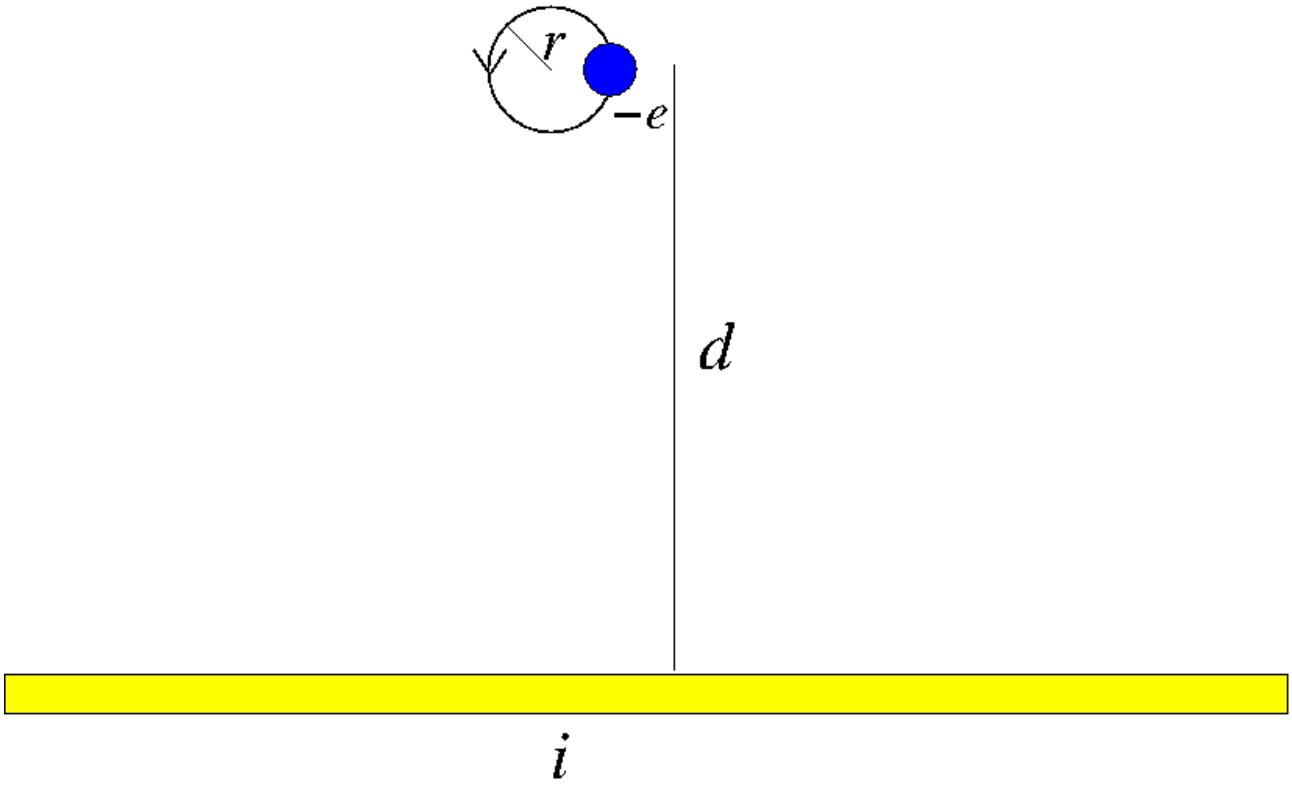
F) $2.5 \cdot 10^{-18}$ N, towards the left

G) $5.0 \cdot 10^{-18}$ N, towards the left

H) $7.4 \cdot 10^{-18}$ N, towards the left

I) Don’t know

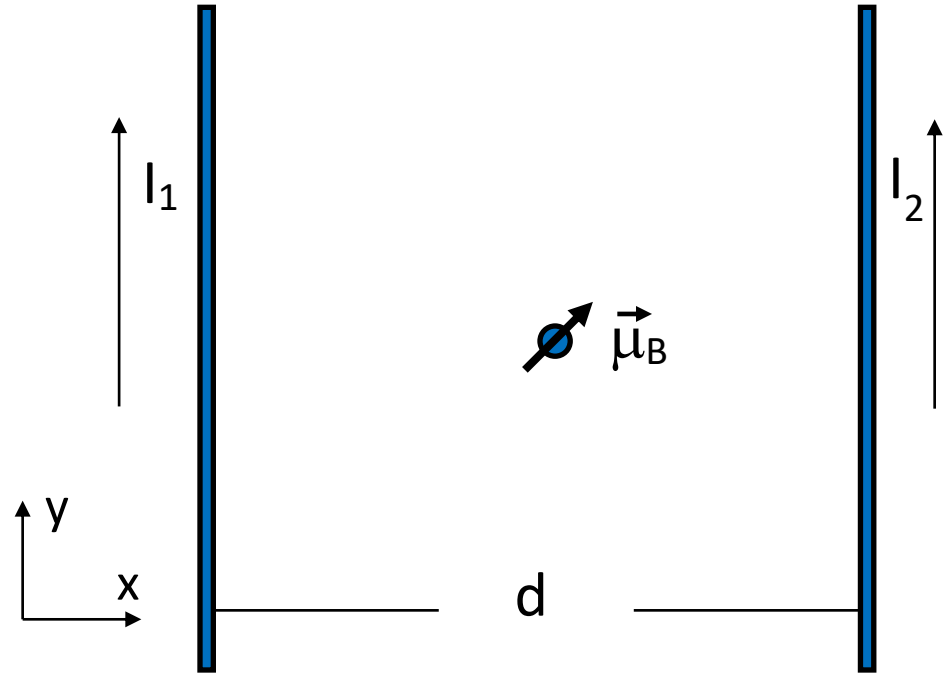
Exercise 5.



An electron moves in a circle in the magnetic field from an infinitely long straight wire, carrying a current *i* as shown in the figure. The electron is at a distance *d*=1.0 cm from the wire, and the radius of the circular path, *r*, is sufficiently small that the magnetic field from the wire can be assumed constant over the extent of the circle. The frequency of the electron's periodic movement is observed to be $1.0 \cdot 10^7$ s^{-1}. Find the magnitude and direction of the current in the wire.

A) *i*=2.8 A, towards the left
B) *i*=2.8 A, towards the right
C) *i*=1.4 A, towards the left
D) *i*=1.4 A, towards the right
E) *i*=17.9 A, towards the left
F) *i*=17.9 A, towards the right
G) Don't know

Exercise 6.



Two infinitely long straight wires are orienter in parallel with a separation of $d = 10.0$ cm. An atom having a magnetic moment of $\mu_B = 9.27 \cdot 10^{-24}$ J/T oriented at an angle of 45 degrees with the *x*-axis (see figure) is situated exactly at the midpoint between the two wires, which carry the currents $I_1 = 3.00$ A and $I_2 = 10.0$ A. Determine the x-component of the torque $\vec{\tau}$ on the atom from the magnetic field generated by the two wires.

A) $\tau_x = 1.84 \cdot 10^{-28}$ Nm

B) $\tau_x = 2.41 \cdot 10^{-28}$ Nm

C) $\tau_x = 3.41 \cdot 10^{-28}$ Nm

D) $\tau_x = -1.84 \cdot 10^{-28}$ Nm

E) $\tau_x = -2.41 \cdot 10^{-28}$ Nm

F) $\tau_x = -3.41 \cdot 10^{-28}$ Nm

G) Don't know